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STATIC EDGE FINDER AND TOOL HEIGHT GAUGE FOR COMPUTER NUMERICAL CONTROL (CNC) MACHINES

Abstract

A static edge finder for finding an edge of a workpiece loaded in a CNC machine, and tool height gauge for finding the height of a tool loaded in a CNC machine.

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Background/Summary

REFERENCE TO PENDING PRIOR PATENT APPLICATION [0001] This patent application claims benefit of pending prior U.S. Provisional Patent Application Ser. No. 62/982,893, filed Feb. 28, 2020 by Steven E. Phillips for STATIC EDGE FINDER AND TOOL HEIGHT GAUGE FOR COMPUTER NUMERICAL CONTROL (CNC) MACHINES (Attorney's Docket No. PHIL-9A PROV), which patent application is hereby incorporated herein by reference.

FIELD OF THE INVENTION

[0002] This invention relates to Computer Numerical Control (CNC) machines in general, and more particularly to edge finders and tool height gauges for use with CNC machines.

BACKGROUND OF THE INVENTION

[0003] CNC machines are electro-mechanical devices which are used to transform a stock piece of material (i.e., a workpiece) into a finished product (e.g., by machining, drilling, etc.). CNC machines typically comprise a work enclosure for isolating the workpiece from the operator (and from bystanders) while the workpiece is worked. Doors allow access into the work enclosure, e.g., for the insertion and removal of workpieces, for supplying tools to the CNC machine and removing tools from the CNC machine, etc.

[0004] For safety reasons, all new CNC machines come with door interlocks that lock the doors of the work enclosure while the spindle of the CNC machine is running, thereby ensuring that operators (and bystanders) are shielded from the cutting action taking place in the work enclosure during operation of the CNC machine. If the CNC machine's safety switches are tampered with, the company could be severely fined.

Finding The Edges Of Workpieces

[0005] In many cases, it is necessary to identify the edges of a workpiece when the workpiece is positioned in the CNC machine. The traditional way of doing this is to use a conventional edge finder **5** (see FIGS. **1** and **2**) that is loaded into the spindle **10** of the CNC machine and rotates with the spindle of the CNC machine. The conventional edge finder **5** generally comprises an upper portion **15** which is loaded into the spindle **10** of the CNC machine, and a lower portion **20** which is normally spring-biased into an eccentric position (i.e., so that the center axis **25** of the lower portion **20** is eccentric to the center axis **27** of the upper portion **15**). While spinning, the edge finder **5** encounters an edge **30** of a workpiece **35** and the lower portion **20** of the edge finder is forced laterally, so that the center axis **25** of the lower portion **20** is moved into alignment with the center axis **27** of the upper portion **15** of the edge finder (and hence into alignment with the center axis **40** of the spindle **10**), thereby indicating to the operator that the edge finder **5** is flush with the edge **30** of the workpiece **35**. At this point the spindle **10** of the CNC machine is raised and the location of the spindle axis **40** is recorded. Half the diameter of the lower portion **20** of edge finder **5** is then added to, or subtracted from, the center axis **40** of the spindle **10** so as to identify the edge **30** of the workpiece **35**. However, with the door of the work enclosure closed, it is very difficult to see this happen, which makes edge detection via a conventional edge finder slow and inconvenient.

[0006] A more modern (but much more expensive) way to locate the edge of the workpiece is to use an electronic spindle probe **45** or other electronic device. See FIGS. **3** and **4**. These are expensive, or may require batteries, and are very delicate.

[0007] As will hereinafter be discussed, the present invention provides a novel static edge finder for finding the edges of workpieces which overcomes the problems associated with the prior art.

Determining the Height of Tools Mounted in the Spindle of a CNC Machine

[0008] In addition to the foregoing, in many cases it may be necessary to determine the height of a tool mounted in the spindle of the CNC machine, so that the tool can properly interface with the workpiece. More particularly, different tools may have different heights, so that the different tools project different distances down from the spindle of the CNC machine. Therefore it is frequently necessary to determine the height of a tool mounted in the spindle of the CNC machine, so that the tool can properly interface with the workpiece.

[0009] A common method for determining the height of the tool mounted in the spindle of the CNC machine involves hand-positioning a gauge block in the workpiece enclosure and then moving the tool to find the top of the gauge block. However, this can only be done with the door of the work enclosure open. If an operator has many tools to reference, it becomes a tedious and time-consuming process.
[0010] There are also electric height gauges that “light up” when the tool touches it. This still may require opening the door of the work enclosure to locate the button at the top of the electric height gauge so that, again, where there are many tools, the process is tedious and time-consuming.
[0011] Many CNC machines come with a very expensive option called a “tool pre-setter” which stores the tool height values electronically. This requires a program to do so.
[0012] As will hereinafter be discussed, the present invention also provides a novel tool height gauge for finding the height of tools mounted in the spindle of a CNC machine which overcomes the problems associated with the prior art.
SUMMARY OF THE INVENTION
[0013] The present invention comprises the provision and use of a new and improved static edge finder.
[0014] In one preferred form of the invention, there is provided a static edge finder comprising: [0015] a body having a distal end and a proximal end; an opening formed in the distal end of the body and extending toward the proximal end of the body, the opening terminating in a shoulder; [0016] a slot formed in the body and communicating with the opening; [0017] a plunger having a distal end and a proximal end, the proximal end of the plunger being disposed in the opening of the body and the distal end of the plunger extending out of the opening of the body; [0018] a spring disposed between the proximal end of the plunger and shoulder of the opening; and [0019] a retaining pin mounted to the plunger and extending through the slot.
[0020] And in one preferred form of the invention, there is provided a method for finding an edge of a workpiece loaded in a CNC machine, the method comprising: [0021] providing a static edge finder comprising: [0022] a body having a distal end and a proximal end; [0023] an opening formed in the distal end of the body and extending toward the proximal end of the body, the opening terminating in a shoulder; [0024] a slot formed in the body and communicating with the opening; [0025] a plunger having a distal end and a proximal end, the proximal end of the plunger being disposed in the opening of the body and the distal end of the plunger extending out of the opening of the body; [0026] a spring disposed between the proximal end of the plunger and shoulder of the opening; and [0027] a retaining pin mounted to the plunger and extending through the slot; [0028] mounting the static edge finder to the spindle of the CNC machine; [0029] positioning the distal end of the plunger against the top face of the workpiece; [0030] lowering the spindle of the CNC machine so as to cause the proximal end of the plunger to move closer to the shoulder of the opening against the power of the spring; [0031] moving the spindle of the CNC machine laterally until the distal end of the plunger moves off the top face of the workpiece and sits in engagement with the edge to be found; and [0032] storing the position of the spindle, and then adding or subtracting half the diameter of the distal end of the plunger so as to find the edge of the workpiece.
[0033] The present invention also comprises the provision and use of a new and improved tool height gauge.
[0034] In one preferred form of the invention, there is provided a tool height gauge comprising: [0035] a body having a distal end and a proximal end; [0036] an opening formed in the distal end of the body and extending toward the proximal end of the body, the opening terminating in a shoulder; [0037] a slot formed in the body and communicating with the opening; [0038] a plunger having a distal end and a proximal end, the proximal end of the plunger being disposed in the opening of the body and the distal end of the plunger extending out of the opening of the body; [0039] a spring disposed between the proximal end of the plunger and shoulder of the opening; and

[0040] a retaining pin mounted to the plunger and extending through the slot.
[0041] And in one preferred form of the invention, there is provided a method for finding the height of a tool loaded in a CNC machine, the method comprising: [0042] providing a tool height gauge comprising: [0043] a body having a distal end and a proximal end; [0044] an opening formed in the distal end of the body and extending toward the proximal end of the body, the opening terminating in a shoulder; [0045] a slot formed in the body and communicating with the opening; [0046] a plunger having a distal end and a proximal end, the proximal end of the plunger being disposed in the opening of the body and the distal end of the plunger extending out of the opening of the body; [0047] a spring disposed between the proximal end of the plunger and shoulder of the opening; and [0048] a retaining pin mounted to the plunger and extending through the slot; [0049] mounting the tool to the spindle of the CNC machine; [0050] moving the spindle of the CNC machine so as to cause the tool to engage the distal end of the plunger of the tool height gauge and cause the proximal end of the plunger to move closer to the shoulder of the opening against the power of the spring; and [0051] raising spindle of the CNC machine so that the distal end of the plunger slips beneath the tool and engages the distal end of the tool, whereby to find the height of the tool.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0052] These and other objects and features of the present invention will be more fully disclosed or rendered obvious by the following detailed description of the preferred embodiments of the invention, which is to be considered together with the accompanying drawings wherein like numbers refer to like parts, and further wherein:
[0053] FIGS. **1** and **2** are schematic views showing a conventional edge finder;
[0054] FIGS. **3** and **4** are schematic views showing a conventional spindle probe;
[0055] FIGS. **5-7** are schematic views showing a new and improved static edge finder formed in accordance with the present invention;
[0056] FIG. **8** is a schematic view showing the new and improved static edge finder of FIGS. **5-7** loaded in the spindle of a CNC machine;
[0057] FIGS. **9-12** are schematic views showing the new and improved static edge finder of FIGS. **5-7** being used to find the edge of a workpiece;
[0058] FIGS. **13-16** are schematic views showing various configurations for the body and/or plunger of the new and improved static edge finder of the present invention;
[0059] FIGS. **17-22** are schematic views showing additional configurations for the plunger of the new and improved static edge finder of the present invention;
[0060] FIGS. **23-29** are schematic views showing still other configurations for the new and improved static edge finder of the present invention;
[0061] FIG. **30** is a schematic view showing a new and improved tool height gauge formed in accordance with the present invention; and
[0062] FIGS. **31-33** are schematic views showing the new and improved tool height gauge of FIG. **30** being used to find the height of a tool loaded into a CNC machine.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
Novel Static Edge Finder
[0063] The novel static edge finder of the present invention is used to locate the edge of a workpiece in a CNC machine with the spindle stationary (i.e., not spinning), which means that the static edge finder can be used with the door of the work enclosure open. Looking next at FIGS. **5-8**, there is shown a static edge finder **100** formed in accordance with the present invention. Static edge finder **100** comprises a main body **105** which is held by the spindle **10** of the CNC machine, a plunger **110**, a spring **115**, and a retaining pin **120**. The body **105** has a proximal end **121** and a

distal end **122**. The body **105** also has an opening **125** (e.g., a hole or bore) at the distal end **122** that is sized to closely slidingly receive the stem **130** of the plunger **110**. The spring **115** sits between the stem **130** of the plunger **110** and a shoulder **135** in opening **125** so as to push the plunger away from the distal end **122** of the body **105**. The body **105** also has a slot **140** which communicates with opening **125**. The retaining pin **120** is mounted to the plunger **110** and extends through the slot **140** so as to provide a limited range of motion for the plunger **110**. As a result of this construction, the plunger **110** is free to slide in and out of the body **105** to a limited extent, with the plunger **110** normally biased into an extended position by spring **115**. Plunger **110** preferably includes an enlarged contact head **145** at the distal end of stem **130**. Enlarged contact head **145** may terminate in a rounded convex configuration to facilitate engagement with a workpiece.

[0064] The novel static edge finder **100** is preferably used as follows.

[0065] Step 1. Put the proximal end **121** of the static edge finder **100** in the chuck or collet of the spindle **10** of the CNC machine. See FIG. **8**.

[0066] Step 2. Using the “jog box” of the CNC machine, bring the plunger **110** into contact with the top face **150** of the workpiece **35** close to the desired edge **30** to be located. See FIG. **9**. Note that this is done with the spindle “off”, which means that the door of the work enclosure may be open, whereby to provide excellent visibility for the operator.

[0067] Step 3. Slowly lower the spindle to compress the spring **115** of the static edge finder **100** without “bottoming out” the retaining pin **120** in the slot **140** of the body **105**. See FIG. **10**. Again, this is done with the spindle **10** of the CNC machine “off”, which means that the door of the work enclosure may be open, whereby to provide excellent visibility for the operator.

[0068] Step 4. Move the spindle of the CNC machine laterally toward the desired edge **30**, whereby to move the static edge finder **100** laterally towards the desired edge **30**, slowly watching for movement of the plunger **110**. See FIG. **11**. This lateral movement is done with the spindle “off”, allowing the door of the work enclosure to be open.

[0069] Step 5. When the edge of the plunger **110** snaps off the top face **150** of the workpiece **35**, so that the plunger **110** sits in engagement with desired edge **30**, store the position of the spindle, then add or subtract half the diameter of the plunger (depending on the direction) so as to identify the position of the edge **30** of the workpiece **35**. See FIG. **12**. Note that where plunger **110** comprises an enlarged contact head **145** at the end of stem **130**, “half the diameter of the plunger” is half the diameter of the enlarged contact head **145**.

[0070] Step 6. Repeat Steps 1-5 for all other remaining edges of the workpiece that need to be located for the various axes of the CNC machine.

Body **105**

[0071] In one preferred form of the invention, the body **105** is round, but it could also be square or a hex, etc., depending on its use. See FIG. **13** (where body **105** is shown with a round configuration) and FIG. **14** (where body **105** is shown with a square configuration).

[0072] The opening (or hole or bore) **125** of the body **105** houses the spring **115** and the stem **130** of the plunger **110**. The slot **140** of the body **105**, in conjunction with the retainer pin **120**, is provided to limit the stroke of the plunger **110**. Preferably, the slot **140** is set parallel to the longitudinal axis of the body **105**. However, the slot **140** could be helical, which would accentuate the motion of the plunger **110**, making it easier for the operator to detect when the plunger **110** is close to falling off the top face **150** of the workpiece. See FIG. **15**. The body **105** can also have graduation markings **155** (FIG. **15**) engraved onto it, or printed onto it, to give reference to the motion of the retainer pin **120** (and hence reference to the motion of the plunger **110**). If desired, body **105** may have a reduced diameter **160** in the region of graduation markings **155**, and/or a portion of enlarged contact head **145** of plunger **110** can ride over reduced diameter **160**/graduation markings **155** so as to provide enhanced visibility of the position of plunger **110** relative to body **105**. See FIG. **16**.

Plunger **110**

[0073] The plunger **110** comprises a stem (or shank) **130** and a hole **165** for receiving the retainer pin **120**. Plunger **110** preferably also comprises an enlarged contact head **145** at the distal end of stem **130**, with a shoulder **170** being disposed between the contact head **145** and the stem (or shank) **130**.

[0074] The geometry of the plunger **110** can range from very simple to more complex, depending on the desired effect. By way of example but not limitation, the distal end of the contact head **145** can be flat or convex, the height of the contact head can be small or large, the shoulder of the contact head can be relatively close to, or relatively far from, the distalmost point of the contact head, etc. See FIGS. **17-22**.

Design Variations

[0075] The simplest and preferred construction for the body **105** is a straight slot **140** and no graduation markings **155**. See FIG. **23**.

[0076] In another form of the invention, the body **105** can be provided with graduation markings **155** (engraved or printed, including colored printing) as shown in FIG. **24**, and the plunger **110** can be provided with a skirt **175** (acting as the enlarged contact head **145**) as shown in FIG. **25**, so as to give the operator an indication that the plunger is about to, or has, fallen off the top face **150** of the workpiece **35**. More particularly, at the top position of the graduation markings **155** of the body **105**, there may be a red ring; at the graduation marking of the body which is indicative of the plunger being close to falling off the top face **150** of the workpiece, the ring might be a yellow ring; and the graduation marking reflective of when the plunger drops off the top face **150** of the workpiece, a green ring on the graduation marking of the body would be revealed. The colors may vary depending on the meaning desired.

[0077] And, if desired, threaded fasteners **180** can be used in place of the retainer pin **120** to retain the plunger in the body. See FIGS. **26-28**.

[0078] Furthermore, if desired, a gas piston mechanism **180** may be used in place of spring **115** to bias the plunger **110** distally away from the end of the body. See, for example, FIG. **29**.

Novel Tool Height Gauge

[0079] The novel tool height gauge of the present invention is used to determine the height of a tool mounted in the spindle of a CNC machine with the spindle stationary (i.e., not spinning), which means that the novel tool height gauge can be used with the door of the work enclosure open.

[0080] More particularly, and looking next at FIG. **30**, there is shown a novel tool height gauge **200** formed in accordance with the present invention. Novel tool height gauge **200** comprises a body **205** (generally analogous to the body **105** of the static edge finder **100**), an opening **225** formed in the body of the tool height gauge **200** (generally analogous to the opening **125** in the body of the static edge finder **100**), a plunger **210** which is spring-biased (via spring **215**) outwardly from the opening **225** in the body of the tool height gauge (generally analogous to the plunger **110** and spring **125** in the body **105** of the static edge finder **100**), a slot **240** (FIG. **31**) in the body of the tool height gauge (generally analogous to the slot **140** in the body of the static edge finder **100**), and a retainer pin **220** (FIG. **31**) which is fixed to the plunger (via a hole **265**) and which extends through the slot **240** in the body of the tool height gauge **200** (in a manner generally analogous to how the retainer pin **120** of the static edge finder **100** extends through the slot **140** in the body of the static edge finder).

[0081] The novel tool height gauge allows the operator to acquire the reference height of a tool **285** mounted in the spindle **10** of a CNC machine with the door of the work enclosure closed by moving the tool **285** into engagement with the distal face **290** of the plunger **210** of the tool height gauge **200** and then shifting the tool **285** laterally toward body **205** so as to force the plunger **210** laterally and compress the spring **215** (see FIG. **31**). Next, the operator will raise the tool **285** up (see FIG. **32**) until the plunger **210** slips underneath the tool **285** (see FIG. **33**). This will indicate to the operator that the tool **285** is positioned at a known height, whereupon the value is stored in the control of the CNC machine by the operator.

[0082] Significantly, these operations can all be undertaken with the door open or closed, so that a plurality of tools can have their heights measured without requiring the door to be opened or closed between tool height measurements. This is an important feature, since CNC machines often use 10-50 (or more) tools in a single workpiece run.

Modifications Of The Preferred Embodiments

[0083] It should be understood that many additional changes in the details, materials, steps and arrangements of parts, which have been herein described and illustrated in order to explain the nature of the present invention, may be made by those skilled in the art while still remaining within the principles and scope of the invention.

Claims

1. A static edge finder for finding an edge of a workpiece loaded in a CNC machine, the static edge finder comprising: a body having a distal end and a proximal end; an opening formed in the distal end of the body and extending toward the proximal end of the body, the opening terminating in a shoulder; a slot formed in the body and communicating with the opening; a plunger having a distal end and a proximal end, the proximal end of the plunger being disposed in the opening of the body and the distal end of the plunger extending out of the opening of the body; a spring disposed between the proximal end of the plunger and shoulder of the opening; and a retaining pin mounted to the plunger and extending through the slot; wherein the static edge finder is configured to be mounted to a spindle of the CNC machine to find the edge of the workpiece loaded in the CNC machine.

2. A tool height gauge for finding the height of a tool loaded in a CNC machine, the tool height gauge comprising: a body having a distal end and a proximal end; an opening formed in the distal end of the body and extending toward the proximal end of the body, the opening terminating in a shoulder; a slot formed in the body and communicating with the opening; a plunger having a distal end and a proximal end, the proximal end of the plunger being disposed in the opening of the body and the distal end of the plunger extending out of the opening of the body; a spring disposed between the proximal end of the plunger and the shoulder of the opening; and a retaining pin mounted to the plunger and extending through the slot; wherein the tool height gauge is configured to be mounted to a spindle of the CNC machine to find the height of the tool loaded in the CNC machine.

3. The static edge finder according to claim 1 wherein the plunger comprises an enlarged contact head.

4. The static edge finder according to claim 3 wherein the enlarged contact head comprises a rounded convex configuration.

5. The static edge finder according to claim 3 wherein the enlarged contact head comprises a flat configuration.

6. The static edge finder according to claim 1 wherein the body comprises a round configuration.

7. The static edge finder according to claim 1 wherein the body comprises a square configuration.

8. The static edge finder according to claim 1 wherein the slot is set parallel to the longitudinal axis of the body.

9. The static edge finder according to claim 1 wherein the slot comprises a helical configuration.

10. The static edge finder according to claim 1 wherein the body comprises graduation markings.

11. The static edge finder according to claim 10 wherein the body comprises a reduced diameter in the region of the graduation markings.

12. The static edge finder according to claim 11 wherein a portion of the plunger is configured to ride over the reduced diameter.

13. The tool height gauge according to claim 2 wherein the body comprises a square configuration.

14. The tool height gauge according to claim 2 wherein the slot comprises a rounded configuration.

15. The tool height gauge according to claim 2 wherein the slot is set parallel to the longitudinal axis of the body.
16. The tool height gauge according to claim 2 wherein the slot comprises a helical configuration.